

NAMPULA, Mozambique

WMO: 672370

Lat: 15.106S

Lon: 39.282E

Elev: 1444

StdP: 13.95

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	60.0	61.1	48.9	54.0	83.8	51.4	59.4	81.2	17.6	70.6	16.4	71.5	7.0	240	0.312

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	21.4	95.4	71.9	94.0	71.9	92.4	71.9	77.4	86.7	77.1	86.4	76.4	85.3	8.1	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	140.8	79.9	74.9	138.0	79.3	73.9	133.3	78.6	42.1	86.5	41.5	86.4	40.9	86.1	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.0	15.4	13.9	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	77.0	80.4	79.6	79.1	76.5	74.5	71.2	70.6	72.8	76.0	79.3	81.7	82.0		
	DBStd	4.53	1.95	2.01	2.05	2.42	2.20	2.58	2.58	2.53	2.68	2.73	2.65	2.54		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	2	0	0	0	0	0	1	1	0	0	0	0	0		
	CDD50	9839	942	830	903	794	760	635	640	705	780	907	951	992		
	CDD65	4367	477	410	438	344	295	186	176	241	330	442	501	527		
	CDH74	34397	3844	3127	3144	2049	1677	740	795	1547	2893	4358	5186	5037		
	CDH80	13331	1461	1135	1022	509	323	73	110	380	1196	2144	2639	2339		
(14) Wind	WSAvg	6.9	5.4	5.7	5.8	6.6	6.3	7.2	7.2	7.2	8.0	8.5	8.2	7.1		
(15) Precipitation	PrecAvg	41.2	10.2	8.4	7.2	3.3	0.9	0.6	0.8	0.3	0.3	1.0	2.8	6.9		
	PrecMax	62.8	21.5	16.0	16.6	9.5	4.8	2.9	2.6	2.1	3.0	3.9	12.1	16.6		
	PrecMin	29.1	0.8	2.6	1.0	0.3	0.0	0.0	0.2	0.0	0.0	0.0	0.2	0.6		
	PrecStd	10.7	5.6	3.5	3.8	2.2	0.9	0.6	0.6	0.5	0.5	1.2	2.6	4.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.1	91.8	90.4	89.6	87.9	85.0	85.9	88.8	93.2	96.8	98.4	97.1		
		MCWB	75.7	76.1	75.5	72.7	69.8	67.0	66.1	66.8	68.4	71.3	71.0	74.2		
	2%	DB	91.2	89.9	88.6	86.5	85.4	82.3	83.0	86.3	91.0	94.0	96.0	95.3		
		MCWB	75.5	75.8	75.4	73.2	70.2	67.1	65.8	66.1	67.5	70.1	70.8	73.5		
	5%	DB	89.4	88.1	87.4	84.8	83.4	80.4	80.8	84.2	89.1	92.3	94.5	93.4		
		MCWB	75.3	75.0	75.3	73.1	69.8	66.7	65.8	65.7	67.4	69.9	71.3	74.0		
	10%	DB	87.6	86.4	85.8	83.0	82.0	78.5	78.7	82.2	86.8	90.4	92.4	91.3		
		MCWB	75.3	74.5	74.9	72.5	69.5	65.9	65.2	65.6	67.3	69.6	71.2	74.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.7	78.3	77.8	76.7	73.8	71.3	71.6	71.1	72.0	75.2	76.2	78.9		
		MCDB	86.7	87.9	85.9	83.9	81.6	78.0	78.7	80.8	84.4	89.8	90.2	89.5		
	2%	WB	77.4	77.2	77.1	75.5	72.6	69.8	68.4	69.2	70.6	73.1	74.7	77.2		
		MCDB	86.3	86.3	85.2	82.4	79.9	76.8	76.5	78.7	83.7	87.4	86.6	88.7		
	5%	WB	76.8	76.6	76.4	74.8	71.8	68.4	67.7	68.2	69.7	71.9	73.7	76.3		
		MCDB	85.9	85.2	84.1	81.4	78.8	75.2	75.6	77.6	82.8	85.3	85.1	87.3		
	10%	WB	75.8	75.6	75.6	73.8	71.0	67.9	66.6	67.3	68.6	71.2	73.1	75.4		
		MCDB	84.4	83.3	82.7	79.5	77.9	74.6	74.2	76.5	80.6	84.2	84.3	85.5		
(35) Mean Daily Temperature Range	MDBR		15.7	15.4	14.3	14.5	16.0	16.0	16.5	19.1	21.9	22.4	21.4	18.4		
	5% DB	MCDBR	17.9	17.4	16.0	16.2	17.6	17.6	19.3	22.4	25.2	25.2	24.5	21.6		
		MCWBR	5.8	5.6	5.1	5.5	5.5	5.7	6.4	6.0	6.3	5.9	5.4	5.2		
	5% WB	MCDBR	16.1	16.6	14.4	13.8	15.7	15.5	17.1	19.5	22.6	23.0	20.8	18.5		
		MCWBR	6.5	5.9	5.7	5.4	6.0	6.1	7.2	6.5	6.9	6.6	5.7	6.1		
(40) Clear-Sky Solar Irradiance	taub		0.403	0.393	0.377	0.354	0.336	0.324	0.329	0.355	0.408	0.441	0.426	0.410		
	taud		2.456	2.489	2.538	2.583	2.590	2.607	2.572	2.453	2.296	2.235	2.310	2.419		
	Ebn at Noon		298	299	297	294	289	287	288	289	284	282	290	296		
	Edh at Noon		38	37	34	31	29	27	29	34	43	47	44	40		
(44) All-Sky Solar Radiation	RadAvg		1851	1864	1807	1683	1538	1386	1399	1630	1905	2041	2062	2013		
	RadStd		137	134	92	71	36	44	57	72	56	82	101	104		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page